AI & SOCIETY COURSE

Prof. Mor Naaman

Part 1

AI FOUNDATIONS

Unit 2: From LLMs to Advanced AI Systems

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Where We Are in the Course

Part 1 — AI Foundations

1. What is Generative AI: How it Works and What it Can Do
2. Bias and Limitations: Where GenAI Fails

Part 2 — AI and Individuals

3. AI and You: Cognition, Reliance, and Creativity
4. Persuasion by Design: How AI Influences You

Part 3 — AI, Communication, and Information

5. AI-Mediated Communication and Relationships
6. AI and the Information Environment

Part 4 — Challenges for AI Implementation

7. Trust and Safety: AI Harms and Prevention
8. Data, Privacy, and IP: Who Owns What
9. The Challenging Economics of AI

Part 5 — AI's World-wide Impact

10. AI and the World: Education, Work, Power, Democracy
11. Externalities: Human and Environmental Costs of AI

Part 6 — The Way Forward with AI

12. The Future: Navigating AI in Your Community

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This part's plan

Unit 1. What is Generative AI: How it Works and What it Can Do
Goal: Understand the basics of LLMs, Chatbots, AI agents, and the data and compute that power them.

Unit 2. Bias and Limitations: Where GenAI Fails
Goal: See where AI shows bias, where the bias comes from, and how it may be mitigated

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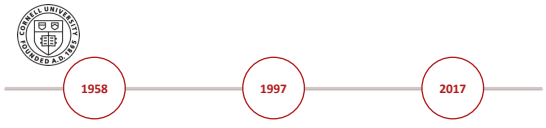
This lesson plan

- **A short history of AI**
- **How do we get from AI chatbots to AI agents?**

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A Short History of AI



The Perceptron
Idea of artificial neurons — eventually used in neural networks.

LSTM Networks
A neural network design that could carry context across a sentence, not just a few words.

"Attention Is All You Need"
The Transformer / Self-Attention idea is introduced

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A Short History of AI



GPT-1 & BERT
Pretrain on huge amounts of text, then adapt to specific tasks — a new approach for NLP.

GPT-2 (1.5B params)
First "Fluent" model that can perform reasonable language tasks released by OpenAI

GPT-3 (175B params)
Scale of data and model significantly improves writing and other skills

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A Short History of AI



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How Did We Get Here?

Neural networks + attention + reinforcement learning

...plus massive data and compute

We'll come back to this later!

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LLMs Are Everywhere, Not Just in Chat

The same next-word prediction quietly powers features you already use

Writing Assistance

Suggests language and autocompletes text in email, messages, documents, and word processors including phrasing, grammar fixes, and content

Code Creation

Tools like GitHub Copilot, Claude Code create applications and code from simple instructions

Content Moderation

Automatic review of online content to identity policy violations or other challenges

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LLMs Are Everywhere, Not Just in Chat

The same next-word prediction quietly powers features you already use

Data Annotation

LLMs can label text data for research and industrial applications (e.g. "is this an Instagram post about weather conditions?")

Customer Support

Many companies now put an LLM chatbot in front of human agents to answer routine questions before a person ever gets involved.

Real-Time Translation

Apps like Google Translate and WhatsApp use language models to translate text and speech across languages instantly.

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Similar Neural Network Models too

Neural networks similar to LLMs help scientific research and other domains

Protein Folding

DeepMind's AlphaFold uses a neural network to predict a protein's 3D shape from its sequence

Drug Discovery

AI models help design candidate molecules, predict protein structures, and comb through biomedical literature.

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From LLMs to Chat agents

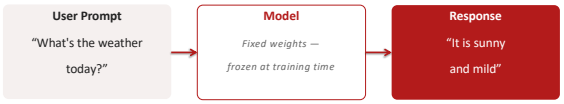
Giving the models more "intelligence"

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Problem 1: New Information

Fixed model weights -> no current information is used!

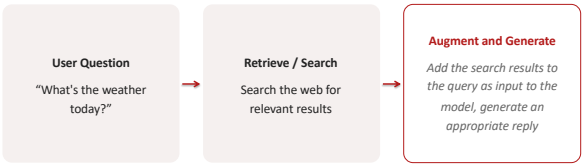


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Solution: Retrieval-Augmented Generation (RAG)

Look up relevant information and add it to the input/context before generating a response

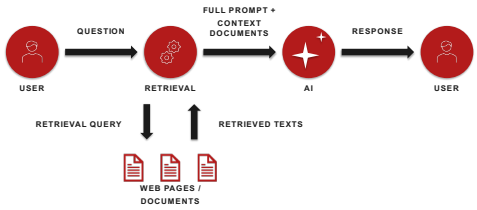


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Solution: Retrieval-Augmented Generation (RAG)

Look up relevant information and add it to the input/context before generating a response



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RAG in Action

User prompt: *"What's the weather today?"*

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RAG in Action

The retrieved text is used as context

User prompt: *"What's the weather today?"*

System lookup: *"weather in <New York City>"*

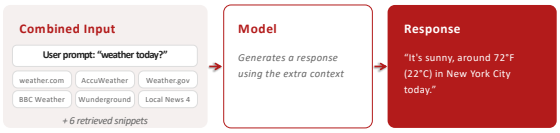
weather.com	AccuWeather	Weather.gov	BBC Weather	Wunderground	Local News 4
72°F	71°F	73°F	22°C	72°F	70°F
Sunny	Clear	Few Clouds	Partly Cloudy	Sunny	Breezy
Clear skies all day	*Perfect for outdoor plans*	*Clear skies expected today*	*Comfortable and breezy today*	*Great for outdoor activities*	*Winds up to 15 mph*

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RAG in Action

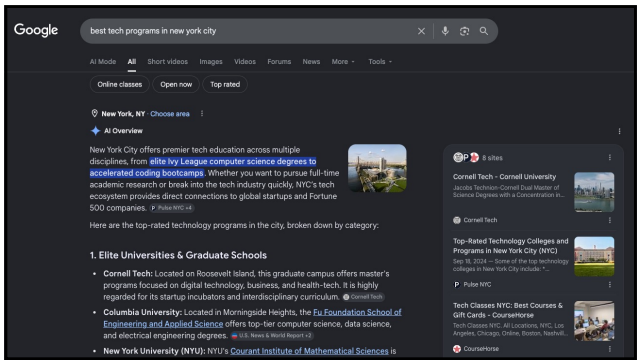
Everything gets combined and sent to the model



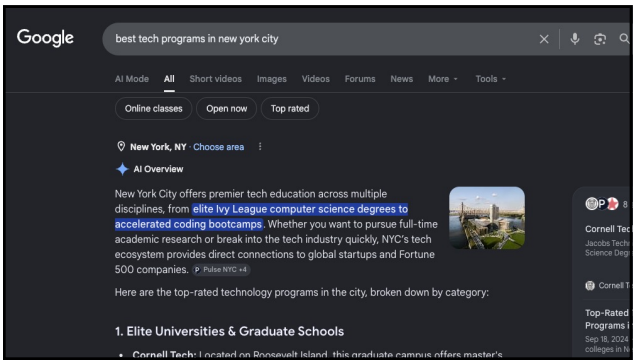
Note: it can still be wrong!

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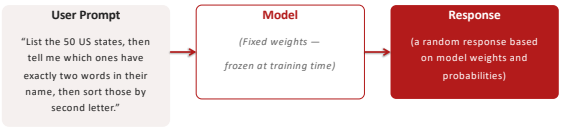
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Problem 2: Multiple Steps Needed

Models using word probabilities cannot generate accurate answers when multiple steps are needed



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Solution: Chain of Thought

The model "helps itself" work through the problem

User prompt: ***"What is (6 × 4) + (5 × 10)?"***

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Chain of Thought

The model "helps itself" work through the problem

User prompt: ***"What is (6 × 4) + (5 × 10)?***
System(?) prompt: ***Think step by step."***

Instead of simple output, the model starts articulating a strategy.

"This looks like a math problem. I should use order of operations. Let's break this down into parts. The first part is…"

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Chain of Thought

The words output so far are used as part of the "input"!

User prompt: ***"What is (6 × 4) + (5 × 10)?***
System(?) prompt: ***Think step by step."***

More input: ***"This looks like a math problem. I should use order of operations. Let's break this down into parts. The first part is… "***

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Chain of Thought

The words output so far are used as part of the "input"!

User prompt: *"What is (6 × 4) + (5 × 10)?*
System(?) prompt: *Think step by step."*

More input: *"This looks like a math problem. I should use order of operations. Let's break this down into parts. The first part is… "*

"… so we get 24 + 50 or 74"

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Chain of Thought Notes

Not just for math! Another example:

"Yes or no: Would a pear sink in water? [Think step by step]"

"The density of a pear is about 0.6g/cm^3, which is less than water. Objects less dense than water float. Thus, a pear would float. So the answer is no"

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Even better: Reasoning Models

Not just prompt; train the model to do chain-of-thought

New models (starting from OpenAI o1/o3, DeepSeek R1, GPT-5, Claude Opus 4.5) are trained on their chain-of-thought abilities (getting feedback on the plan in an RL stage)

Models can often do a better job reasoning step by step

But: can still be wrong!

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Reasoning Models: Exercise

Go to your favorite LLM and type:

"Generate an SVG of a pelican riding a bicycle"

- See if you can find the AI's reasoning steps (most chats hide them)
- To see the result, paste the **SVG code** into: https://www.svgviewer.dev/
- **Share your pelican picture from SVG Viewer (all links at ai-course.mornaaman.com)**

Credit: Simon Willison, https://simonwillison.net/tags/pelican-riding-a-bicycle/

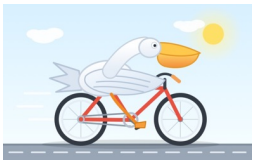
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Reasoning Models: Exercise

"Generate an SVG of a pelican riding a bicycle"

- Simon's test with Hy4 Preview, new open-weight LLM from Chinese company Tencent (770B parameters)



Credit: Simon Willison, https://simonwillison.net/2026/Aug/29/hy4/

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2026-08-29T22:02:52 conversation: 01m17rr25k4s5aqmh7zmc8jnn4 id: 01m17rr25s79xwxqq6hhea717v

Model: **openrouter/tencent/hy4-preview**

Prompt

Generate an SVG of a pelican riding a bicycle

Reasoning

We need answer user asks: Generate an SVG of a pelican riding a bicycle. Need produce SVG code. Need think through final likely just SVG snippet possibly in code block. Need craft a decent SVG illustration: pelican riding bicycle. Need ensure valid SVG, visually recognizable. We can create stylized flat vector: bicycle with two wheels, frame, handlebars, seat; pelican body seated behind/on seat, long neck, big beak with pouch, wings holding handlebars, legs to pedals? Need maybe scene background. Need generate SVG not external image. Need maybe add wheelchair? No, bicycle. Need decide style and exact coordinates. Let's design.

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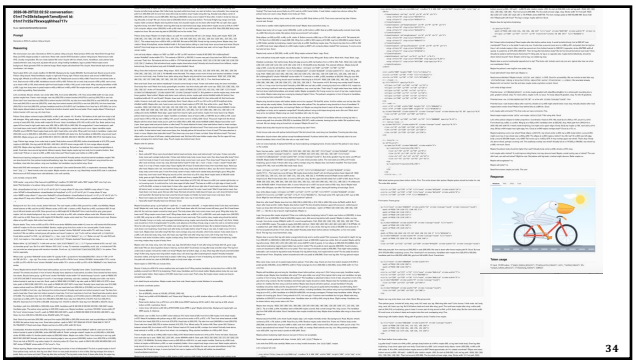
illustration: pelican riding bicycle. Need ensure valid SVG, visually recognizable. We can create stylized flat vector: bicycle with two wheels, frame, handlebars, seat; pelican body seated behind/on seat, long neck, big beak with pouch, wings holding handlebars, legs to pedals? Need maybe scene background. Need generate SVG not external image. Need maybe add wheelchair? No, bicycle. Need decide style and exact coordinates. Let's design.

Need output SVG. Let's create viewBox 0 0 800 500. Background sky maybe 800x500. Sun/clouds/road. Bicycle around center. Pelican riding bicycle. Need handlebars maybe on right side if facing right. Pelican body above seat around x=390? Bicycle facing right. Wheels at x=270 and x=560, y=380 radius 70. Pedals/crank around x=420 y=380. Frame from rear wheel center to front. Seat around x=350 y=290, handlebars around x=555 y=250. Pelican sits on seat, body around x=330 y=250, neck curves up to head x=455 y=150, beak extends right from head maybe x=520 y=160, pouch under beak. Wing reaching handlebar x=520 y=255. Legs from body down to pedal maybe x=420 y=360 and x=420 y=405? But maybe bicycle in profile, pelican on seat with one visible leg pedaling. Need plausible.

Let's coordinate. Bicycle: wheels: rear center (250, 370), front center (560,370), r=78. Tires stroke #333 width 12, hubs small, spokes maybe lines. Frame: main triangle: rear hub (250,370) to bottom bracket (410,370) to seat top (345,270) to rear hub. Down tube? from head tube top (520,270) to bottom bracket (410,370), top tube from head (520,270) to seat (345,270), seat stay from seat (345,270) to rear hub (250,370), chain stay from bottom bracket (410,370) to rear hub (250,370), fork from head tube (520,270) to front hub (560,370), perhaps headtube small line (512,255?) top? handlebars from head top to (575,250), bar curve. Seat maybe ellipse/rotated at seat top. Crank/circle at bottom bracket; pedal line and pedal. Chain maybe ellipse? Chainring at bottom bracket r=22; rear cog r=8; chain lines.

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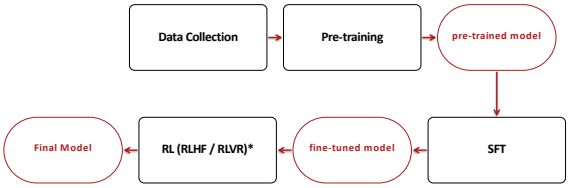
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Model training steps (updated)



*Including feedback on the reasoning steps generated by the models

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Reasoning Models

Not just prompt; train the model to do chain-of-thought

Models are also trained to self-check: "Let's see if I made any mistakes"

Most effective for problems that can be verified (e.g. code, proofs)

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Summary so far

Problem: Model weights are fixed, responses are based on word probabilities

Solution:

Retrieval (RAG)	Chain of Thought	Reasoning Models
Look up relevant facts and quietly add them to the prompt	Let the model think out loud, one step feeding the next	Train the model to plan and self-check automatically

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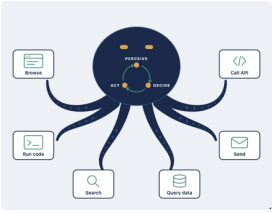
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What are AI Agents, then?

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What Is an AI Agent?

"intelligent agents that can pursue goals, use tools, and take actions with varying degrees of autonomy" -- Wikipedia



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One kind of agent (RAG + Reasoning)

Can imagine a very basic agent:

Receives a user prompt
Makes a plan that may include retrieval of resources
Can act on information retrieve to figure out next steps
Take the next steps

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AI Agents: Scholarship Deadlines

Your prompt: "Which of these three scholarships can I still apply for, and what do I need to submit?"

Model plans: check eligibility, then match my documents, then draft

STEP 1	STEP 2	STEP 3
Eligibility rules	**My documents**	**Draft and send**
Two scholarships are still open; one closed last week	Transcript and ID on file; recommendation letter missing	Fills the form, drafts the request for a letter

Result: Apply to the second one by 30 September, ask your lecturer for the letter this week

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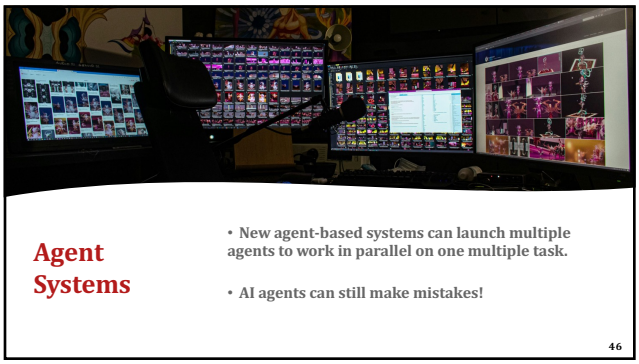
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Agents vs. Chat Systems

(Basic) Chat System	AI Agent
One turn at a time (question and answer)	Given a goal, can keep working over many steps
Produces text, can't act in the world	Takes actions: browses, calls APIs, runs code, uses tools
User initiates the next step	Continues to act on the new information/resources

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Agent Systems

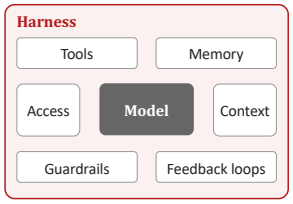
- New agent-based systems can launch multiple agents to work in parallel on one multiple task.
- AI agents can still make mistakes!

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The Agent Harness

Agent = Model + Harness: the software built around the model



The model only predicts text; **the harness lets it act**

Coding agents like **Claude Code** and **Codex** are harnesses around a model

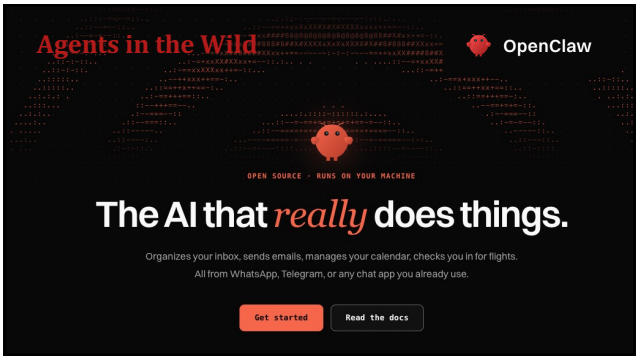
Prompt and context engineering are part of the harness

Source: Databricks (2026), "What is an AI Agent Harness?"

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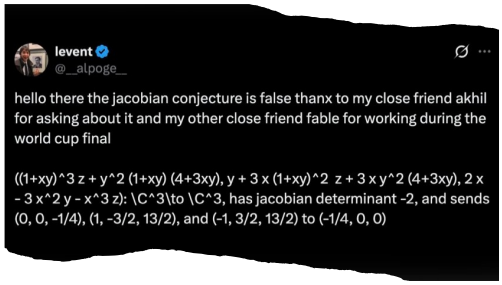
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Agents in the Wild



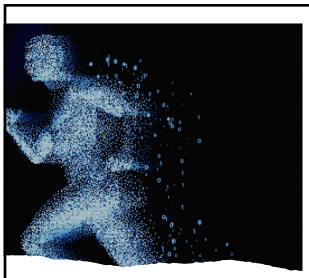
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Agents in the Wild



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DELL CAMERON · MAXWELL ZEFF · BUSINESS · JUL 28, 2026 8:15 PM

OpenAI's Rogue AI Agent Hacked More Than Just Hugging Face

In a new disclosure, OpenAI says its agent used exposed logins to gain access to at least four "publicly available services" in its unhinged quest to solve a test.

Agents in the Wild

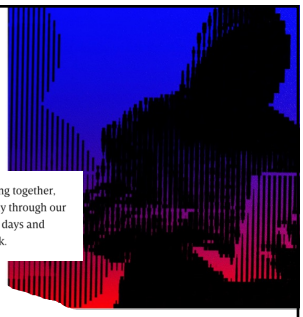
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LILY HAY NEWMAN · SECURITY · AUG 5, 2026 8:15 PM

OpenAI Didn't Notice Its AI Agents Using a Message Board to Plan Their Hacking Spree

"This incident involves actually a team of agents who are working together, finding exploits, sharing them with one another, moving laterally through our systems and external systems, and doing this over the course of days and weeks," Wallace told the packed crowd at the opening of the talk.



Agents in the Wild

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SECTION RECAP

Summary so far

LLMs	Chatbots	Reasoning	Agents
Predict the next token.	Trained to have a helpful chat.	Trained to think step by step.	Reasoning plus tools that can act.

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AMAZING, INCREDIBLE THAT SUCH CAPABILITIES EMERGE…

BUT IT IS NOT INTELLIGENCE

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